ReviewRadar – *Web platform for analyzing customer feedback using AI models.*

Checkpoint 1 - MOPS Laboratory Project

Team and Scrum roles:

Member	Role
Cosmin Mărășoiu	Scrum Master + Developer
Alexandru Zamfir	Product Owner
Delia Tapai	Developer
Horia Zafiu	Developer

1. Product Goal

ReviewRadar helps product teams understand customer feedback faster. Users import reviews, analyze sentiment, identify similar feedback and track issues that come up repeatedly.

MVP goal: CSV import, review validation, automated sentiment analysis and a dashboard presenting the results.

2. Requirements Document

Problem: Customer feedback comes from different sources. Reviewing it manually takes time, and comments describing the same issue can be overlooked. ReviewRadar collects reviews in one place and provides a better understanding of feedback trends.

Users:

User type	Responsibilities
Analyst	Imports, reviews, searches, and filters customer feedback
Product Manager	Monitors results and recurring issues
Administrator	Manages access, projects, and categories

Functional requirements:

ID	Requirement	Scope
FR01	User authentication and project access	MVP
FR02	Create and manage feedback projects	MVP
FR03	CSV import with row-by-row validation	MVP
FR04	Detect exact duplicates during import	MVP
FR05	List, search, and filter reviews	MVP
FR06	Start automated sentiment analysis	MVP
FR07	Store estimated sentiment for each review	MVP
FR08	Dashboard with sentiment distribution	MVP
FR09	Project roles and permissions	Extended
FR10	Add reviews manually	Extended
FR11	Generate embeddings and group similar reviews	Extended
FR12	Correct AI predictions and retry failed processing	Extended
FR13	Trends and recurring-issue detection	Extended
FR14	Create and update issues	Extended
FR15	Export results and view correction history	Extended

Non-functional requirements:

ID	Category	Requirement
NFR01	Security	Project data is accessible only to authorized users
NFR02	Validation	Imported files and fields are validated with clear error messages
NFR03	Performance	AI analysis runs asynchronously without blocking the interface
NFR04	Reliability	Processing errors are shown to the user. Importing the same file twice does not create duplicate reviews
NFR05	AI transparency	Predictions are presented as estimates and can be corrected
NFR06	Testability	Business rules and APIs are verified with automated tests
NFR07	Usability	The interface provides simple navigation, filters, and clear action feedback
NFR08	Compatibility	The interface works in modern desktop browsers

Technology stack:

Technology	Role in ReviewRadar
Streamlit	Web interface, CSV uploads, review tables, interactive filters, and analytics dashboard
FastAPI	Backend APIs, authentication and authorization, CSV processing, analysis jobs, and integration with Hugging Face models
PostgreSQL	Relational database for users, projects, reviews, analysis results, issues, and audit records
pytest	Automated unit and API tests, CSV validation tests, business-rule tests, and mocked AI-inference tests

3. MVP workflow

